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Module 2 — Foundations of Physics

This module covers the fundamental building blocks: units, quantities, and vector analysis that underpin all of A-Level Physics.

2.1 SI Base Units

SI Units (Système International): The internationally agreed system of units used in science. All physical quantities can be expressed in terms of seven base units.

Base Quantity	SI Base Unit	Symbol
Mass	kilogram	kg
Length	metre	m
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

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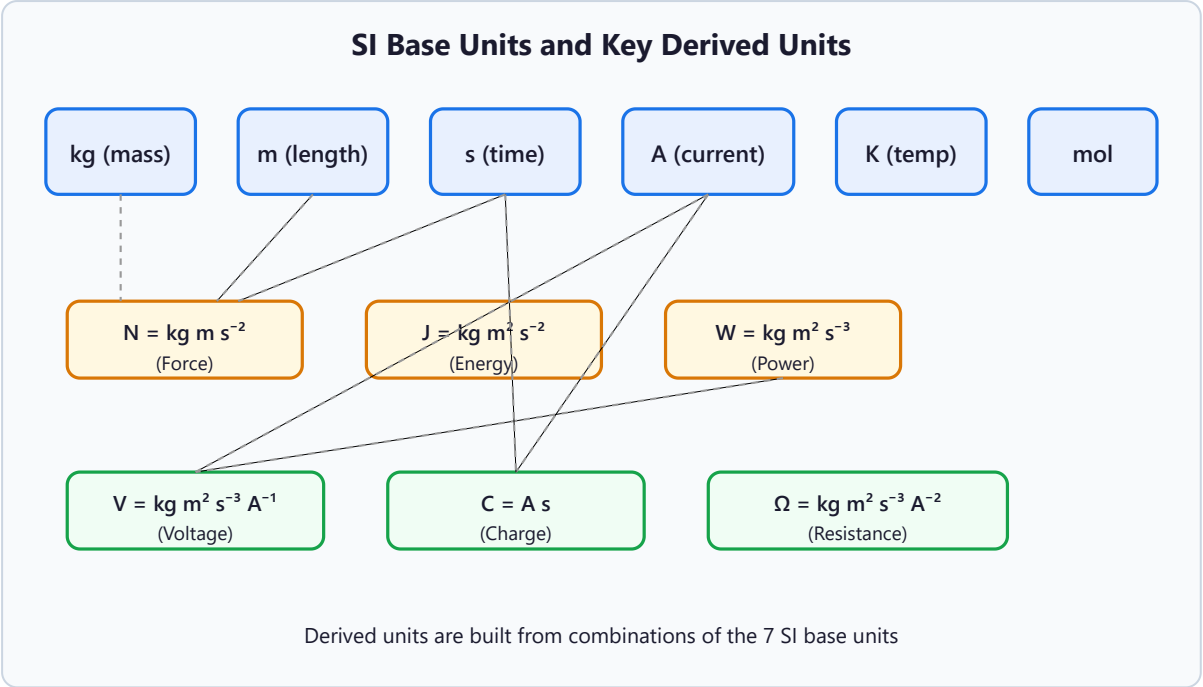

Derived Units

Derived units are combinations of base units. Key examples:

Quantity	Derived Unit	Symbol	In Base Units
Force	newton	N	$kg\ m\ s^{-2}$
Energy / Work	joule	J	$kg\ m^2\ s^{-2}$
Power	watt	W	$kg\ m^2\ s^{-3}$
Pressure	pascal	Pa	$kg\ m^{-1}\ s^{-2}$
Charge	coulomb	C	A s
Potential difference	volt	V	$kg\ m^2\ s^{-3}\ A^{-1}$
Resistance	ohm	Ω	$kg\ m^2\ s^{-3}\ A^{-2}$
Frequency	hertz	Hz	s^{-1}
Magnetic flux density	tesla	T	$kg\ s^{-2}\ A^{-1}$

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Magnetic flux	weber	Wb	$kg\ m^2\ s^{-2}\ A^{-1}$
Capacitance	farad	F	$kg^{-1}\ m^{-2}\ s^4\ A^2$

Exam Tip: To find the base units of a quantity, substitute into the defining equation. For example, for the volt: $V = W/Q = (kg\ m^2\ s^{-2}) / (A\ s) = kg\ m^2\ s^{-3}\ A^{-1}$.

2.2 Homogeneity of Equations

Homogeneous equation: An equation is homogeneous if every term has the same SI base units. This is a necessary (but not sufficient) condition for an equation to be correct.

To check homogeneity:

1. Write out the SI base units of each term in the equation
2. Simplify each term
3. Check that all terms have identical base units

Example: Check $v^2 = u^2 + 2as$

- v^2: $(m\ s^{-1})^2 = m^2\ s^{-2}$
- u^2: $(m\ s^{-1})^2 = m^2\ s^{-2}$
- $2as$: $(m\ s^{-2})(m) = m^2\ s^{-2}$

All terms have units $m^2\ s^{-2}$, so the equation is homogeneous.

Exam Tip: Homogeneity cannot verify dimensionless constants (like the "2" in 2as) or confirm that an equation is physically correct — only that the units are consistent.

2.3 SI Prefixes

Prefix	Symbol	Factor	Standard Form
tera	T	1 000 000 000 000	10^{12}
giga	G	1 000 000 000	10^{9}
mega	M	1 000 000	10^{6}
kilo	k	1 000	10^{3}
centi	c	0.01	10^{-2}
milli	m	0.001	10^{-3}
micro	μ	0.000 001	10^{-6}
nano	n	0.000 000 001	10^{-9}
pico	p	0.000 000 000 001	10^{-12}
femto	f	0.000 000 000 000 001	10^{-15}

Exam Tip: Always convert to SI base units before substituting into equations. Common conversions: 1 km = 10^3 m, 1 nm = 10^{-9} m, 1 μA = 10^{-6} A, 1 kWh = 3.6×10^6 J.

2.4 Estimation and Orders of Magnitude

Order of magnitude: The power of ten closest to a quantity. Two quantities are of the same order of magnitude if they are within a factor of 10 of each other.

Useful Estimates

Quantity	Approximate Value	Order of Magnitude

Diameter of an atom	1×10^{-10} m	10^{-10} m
Diameter of a nucleus	1×10^{-15} m	10^{-15} m
Width of a human hair	~0.1 mm = 10^{-4} m	10^{-4} m
Height of a person	~1.7 m	10^{0} m
Mass of a person	~70 kg	10^{2} kg
Mass of a car	~1500 kg	10^{3} kg
Mass of Earth	6×10^{24} kg	10^{25} kg
Speed of sound in air	~340 m s^{-1}	10^{2} m s^{-1}
Speed of light	3×10^{8} m s^{-1}	10^{8} m s^{-1}
Radius of Earth	6.4×10^{6} m	10^{7} m
Distance to the Sun	1.5×10^{11} m	10^{11} m
Wavelength of visible light	400–700 nm	10^{-7} m

2.5 Scalars and Vectors

Scalar: A quantity that has magnitude only. Examples: mass, speed, distance, energy, temperature, time, power.
Vector: A quantity that has both magnitude and direction. Examples: force, velocity, displacement, acceleration, momentum, electric field strength.

Scalar Quantities	Vector Quantities
Distance	Displacement

Speed	Velocity
Mass	Weight (force)
Energy	Momentum
Temperature	Acceleration
Power	Electric field strength
Pressure	Gravitational field strength

Adding Vectors

Vectors are added using the **triangle rule** or **parallelogram rule**.

Triangle Rule:

1. Draw the first vector as an arrow (to scale)
2. Draw the second vector starting from the tip of the first
3. The resultant vector goes from the start of the first to the tip of the second

Parallelogram Rule:

1. Draw both vectors starting from the same point (to scale)
2. Complete the parallelogram
3. The resultant is the diagonal from the starting point

Special cases for two vectors A and B:

- Same direction: resultant = $A + B$
- Opposite directions: resultant = $|A - B|$
- At right angles: resultant = $\sqrt{(A^2 + B^2)}$, angle $\theta = \tan^{-1}(B/A)$
- At angle θ between them: use the cosine rule or scale drawing

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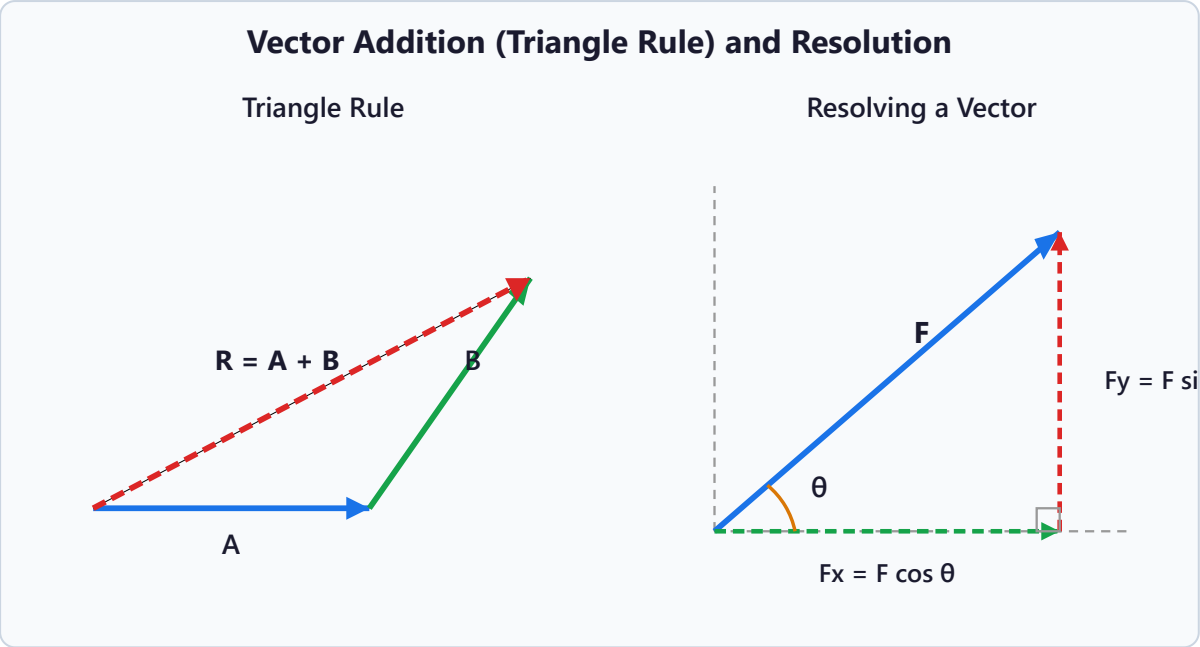

Resolving Vectors into Components

Any vector can be resolved into two perpendicular components. For a vector **F** at angle θ to the horizontal:

$$F_x = F \cos \theta \quad \text{(horizontal component)}$$

$$F_y = F \sin \theta \quad \text{(vertical component)}$$

- The component adjacent to the angle uses cosine
- The component opposite to the angle uses sine
- The original vector can be recovered: $F = \sqrt{(F_x^2 + F_y^2)}$
- The angle can be recovered: $\theta = \tan^{-1}(F_y / F_x)$

Exam Tip: When resolving vectors, always draw a clear diagram showing the angle and label both components. A common mistake is to swap sine and cosine — always check which component is adjacent and which is opposite to the given angle.

Equilibrium

An object is in equilibrium when the resultant force is zero. This means:

- The sum of all horizontal components = 0
- The sum of all vertical components = 0
- For three forces in equilibrium, they form a closed triangle when drawn tip-to-tail

Coplanar forces: Forces that all act in the same plane. For equilibrium of coplanar forces, resolve in two perpendicular directions and set each sum to zero.

Worked Example: A 5.0 N force acts at 30° above the horizontal. Find its horizontal and vertical components.

$F_x = F \cos \theta = 5.0 \times \cos 30° = 5.0 \times 0.866 =$ **4.3 N**

$F_y = F \sin \theta = 5.0 \times \sin 30° = 5.0 \times 0.500 =$ **2.5 N**

Check: $F = \sqrt{(4.3^2 + 2.5^2)} = \sqrt{(18.5 + 6.25)} = \sqrt{24.75} = 5.0$ N ✓

Worked Example: Show that the base units of the pascal (Pa) are kg m^{-1} s^{-2}.

Pressure = Force / Area

Force has units N = kg m s^{-2}

Area has units m^2

So Pa = kg m s^{-2} / m^2 = **kg m^{-1} s^{-2}** ✓